EE 210L: Digital Circuits and Logic Design Lab

Lab 8: Decoders and Encoders

Name: _____ Date: _____

Partner(s): _____ Section: _____

Both chips in this lab use active-low outputs. Record the actual measured voltage and then state the logic level.

Part 5.1: Decoder Truth Table

A2	A1	A0	Output that went LOW	Its measured V (V)	A HIGH output's V (V)
0	0	1			
0	1	1			
1	0	0			

Part 5.2: A Function from Decoder Outputs

F(A2,A1,A0) = Σm(1, 2, 4, 7)

Draw the gate arrangement you used to combine the four active-low decoder outputs. Label every pin.

Figure 1: Combining decoder outputs to form F.

A2	A1	A0	Measured V_{out} (V)	Logic Output	Expected F
0	0	1			
0	1	1			
1	0	0			

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Part 5.3: Priority Encoder Basics

Input held LOW	Ā2	Ā1	Ā0	Decimal index (after inverting)	ḠS	ĒO
None (all HIGH)						
Ī0						
Ī3						
Ī7						

With no input requesting, what distinguishes the output code from a genuine request on input 0?

Part 5.4: Priority Resolution

Inputs held LOW	Ā2	Ā1	Ā0	Decimal index reported	ḠS
Ī2 only					
Ī2 and Ī5					
Ī2, Ī5 and Ī7					
Ī2 and Ī5 (Ī7 released)					

Which input wins when several request at once, and why is that the behavior an interrupt controller needs?

Discussion and Conclusions

Address at least: why active-low outputs lead to a NAND rather than an OR in Part 5.2, what ḠS is for, and how ĒO would let you cascade two encoders into a 16-input priority encoder.

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